

# Jongmin Sung, PhD

AI Research Scientist, Bio AI & Drug Discovery

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## Professional Summary

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**Bio AI Scientist and ML Systems Architect** whose 15+ year career converges on computational drug discovery — from an Applied Physics PhD at Stanford and single-molecule biophysics, through production ML at scale at Roche, to modern agentic AI and protein modeling. Combines wet-lab biology, production ML, and LLM/agent systems in one stack that few scientists hold. Currently leading a cardiac drug discovery collaboration with Stanford researchers, building scientific agents that integrate disease biology, protein structure, compounds, literature, and functional assays. Focused on drug discovery, protein design, biomedical foundation models, and scientific AI agents.

## Core Expertise

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Bio AI and drug discovery | Agentic AI workflows | Genomics and precision healthcare | Biomedical foundation models | Protein modeling and protein language models | Scientific reasoning and LLM evaluation | Production ML and MLOps

## Professional Experience

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**Stanford Cardiovascular Institute (CVI) — Lead AI Research Scientist, Stanford, CA** 06/2026 – Present

- Lead a cardiac drug discovery collaboration with Stanford cardiac researchers and cardiologists.
- Build scientific agents integrating disease biology, protein structures, compounds, literature, and functional assays.
- Develop protein modeling and ligand prediction pipelines for candidate prioritization and experimental validation.

**Sailplane — Senior Knowledge Engineer, San Francisco, CA** 02/2026 – 06/2026

- Built structured knowledge systems for hierarchical AI agents, translating complex operational workflows into machine-readable representations for planning, validation, and decision support.

**Addition Therapeutics — ML Protein Engineer (Consultant), South San Francisco, CA** 07/2025 – 03/2026

- Advanced PRINT genomic medicine platforms through AI-guided protein engineering for precise genome insertion.
- Built scalable, production-ready few-shot active learning ML pipelines using protein language models from scratch.

**Aten Security — Applied AI Research Engineer (Consultant), San Francisco, CA** 07/2025 – 01/2026

- Designed and deployed LLM-powered systems for secure, real-time communication in highly trusted environments.
- Built AI pipelines integrating security-first design, safety constraints, backend services, and cloud infrastructure.

**Scale AI — Human Frontier Collective Specialist, GenAI (Part-time)** 03/2025 – Present

- Evaluate frontier models on advanced scientific, coding, multimodal, and reasoning tasks.

**Roche Sequencing Solutions — Principal Data Scientist, Santa Clara, CA** 10/2019 – 05/2025

- Built SBX sequencing platform, contributing to a Guinness World Record human genome sequencing in 4 hours.
- Applied biophysics, data science, and deep learning to improve next-generation sequencing performance.
- Developed sequencing software pipelines, driving gains in accuracy, read length, and throughput.

**UCSF — Postdoctoral Research Fellow, San Francisco, CA (Advisor: Ron Vale)** 06/2014 – 10/2019

- Developed ML-driven analysis pipelines to model APC cell-cycle regulation mechanisms.
- Characterized human axonemal dynein motility through DNA origami-based in vitro reconstitution.

**Stanford University — Research Assistant, Stanford, CA (Advisor: James Spudich)** 04/2008 – 05/2014

- Invented harmonic force spectroscopy to reveal the force-velocity relationship in muscle contraction.
- Conducted single-molecule studies on human  $\beta$ -cardiac myosin to investigate hypertrophic cardiomyopathy.

## Education

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- **PhD / MS in Applied Physics** | Stanford University, Stanford, CA
- **BS in Physics** | POSTECH, Pohang, South Korea

## Technical Skills

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- **Bio AI & Scientific Computing:** Drug discovery, Protein engineering, Genomics, Transcriptomics, AlphaFold, Boltz-2, PyMOL, BioPython, Statistical modeling, Bayesian inference, Monte Carlo simulation, Signal processing
- **AI & ML:** PyTorch, TensorFlow, scikit-learn, Transformers, LLMs, PLMs, ESM, GNNs, self-supervised learning, few-shot learning, MIL, prompt and context engineering
- **AI Engineering & MLOps:** Claude Code, Harness, agentic coding, hierarchical agent planning, LLM evaluation, GCP, AWS, Azure, Docker, Terraform, CI/CD
- **Programming & Data:** Python, NumPy, Pandas, Go, SQL, Git, Linux, Bash, backend development, API design

## Selected Projects

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- **JSAgent:** AI agent for hard mathematical optimization, evaluated against the Einstein Arena benchmark of unsolved problems in packing, autocorrelation, sphere geometry, and additive combinatorics. (2026) [GitHub]
- **Precision Health Agents:** Multi-agent systems for precision health. (2026) [GitHub]
- **Pancreatic Cancer Survival Prediction:** Predicted patient outcomes using self-supervised learning and multiple instance learning frameworks, integrating TCGA and external datasets. (2025) [GitHub]

## Selected Advisory

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- **Bio-AI / healthcare advisory:** Korean hospital AI strategy & clinical adoption (Valley Innovation Partners, 2026–present); early-stage Bio-AI & therapeutics scouting (NanoStar VC, 2026–present); US market & digital-therapeutics R&D for health-care startups (KITECH & KEIT, 2020–21).

## Selected Publications

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- **Sung, J.** (2026). *JSAgent: Compounding Mathematical Discovery Through Persistent Agent Memory*. arXiv (submitted).
- Hartooni, N., **Sung, J.**, Jain, A., & Morgan, D. O. (2022). Single-molecule analysis of specificity and multivalency in binding of short linear substrate motifs to the APC/C. *Nature Communications*, 13(1), 1–13.
- Niekamp, S., **Sung, J.**, Huynh, W., Bhabha, G., Vale, R., & Stuurman, N. (2019). Nanometer-accuracy distance measurements between fluorophores at the single-molecule level. *PNAS*, 116(10), 4275–4284.
- **Sung, J.**, et al. (2015). Harmonic force spectroscopy measures load-dependent kinetics of individual human  $\beta$ -cardiac myosin molecules. *Nature Communications*, 6(1), 7931.
- Mortensen, K. I.\*, **Sung, J.\***, Flyvbjerg, H., & Spudich, J. A. (2015). Optimized measurements of separations and angles between intra-molecular fluorescent markers. *Nature Communications*, 6(1), 8621. (\* equal contributions)
- Sommesse, R. F.\*, **Sung, J.\***, et al. (2013). Molecular consequences of the R453C hypertrophic cardiomyopathy mutation on human  $\beta$ -cardiac myosin motor function. *PNAS*, 110(31), 12607–12612. (\* equal contributions)

## Awards & Honors

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- 1st place for innovation (among 45 teams from 21 countries), RocheHack Hackathon — 2020
- NRSA/F32 Postdoctoral Fellowship, National Institutes of Health — 2015–2018
- Bio-X Graduate Student Fellowship, Stanford University — 2009–2012